1-D Arrays

Click to add text

Array

- Many applications require multiple data items that have common characteristics
  - In mathematics, we often express such groups of data items in indexed form:
    - $x_1, x_2, x_3, \ldots, x_n$
- Array is a data structure which can represent a collection of data items which have the same data type (float/int/char/…)

Example: Printing Numbers in Reverse

3 numbers

```
int a, b, c;
scanf("%d", &a);
scanf("%d", &b);
scanf("%d", &c);
printf("%d  ", c);
printf("%d  ", b);
printf("%d  \n", a);
```

4 numbers

```
int a, b, c, d;
scanf("%d", &a);
scanf("%d", &b);
scanf("%d", &c);
scanf("%d", &d);
printf("%d  ", d);
printf("%d  ", c);
printf("%d  ", b);
printf("%d  \n", a);
```

The Problem

- Suppose we have 10 numbers to handle
- Or 20
- Or 100
- Where do we store the numbers ? Use 100 variables ??
- How to tackle this problem?
- Solution:
  - Use arrays

Printing in Reverse Using Arrays

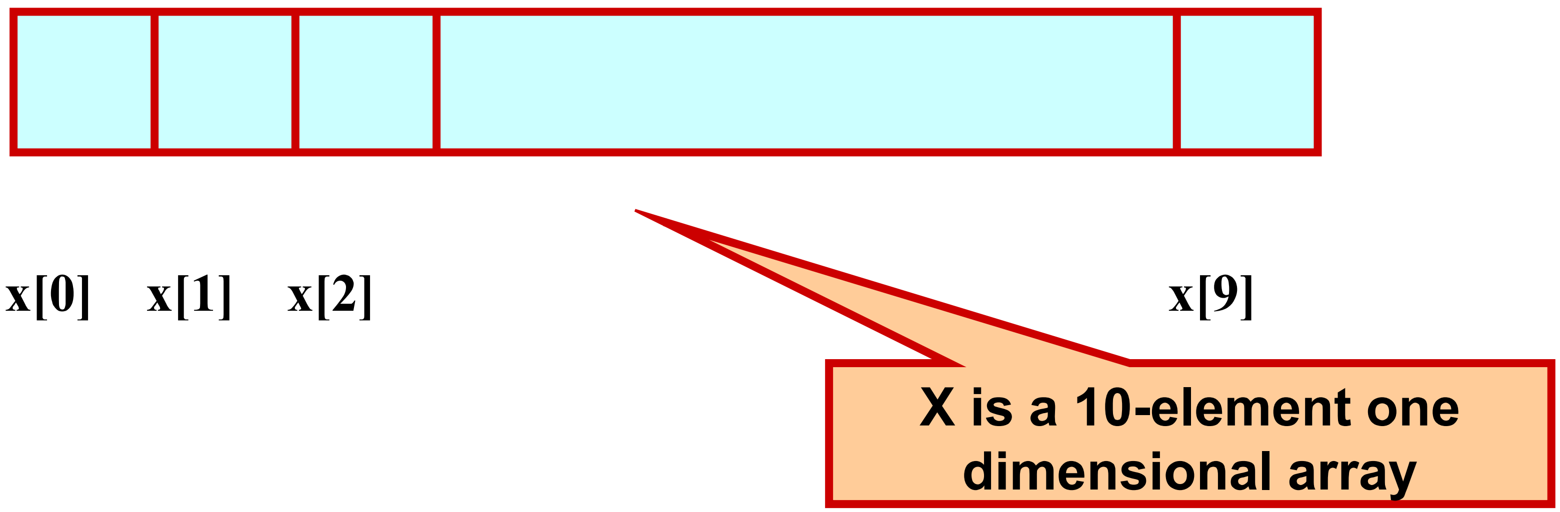
```
int main()
{
    int n, A[100], i;
    printf("How many numbers to read? ");
    scanf("%d", &n);
    for (i = 0; i < n; ++i)
        scanf("%d", &A[i]);
    for (i = n -1; i >= 0; --i)
        printf("%d  ", A[i]);
    printf("\n");
    return 0;
}
```

Using Arrays

- All the data items constituting the group share the same name

    int x[10];

- Individual elements are accessed by specifying the index



First example

```
int main()
{
  int i;
  int data[10];
  for (i=0; i<10; i++) data[i]= i;
  i=0;
  while (i<10)
  {
    printf("Data[%d] = %d\n", i, data[i]);
    i++;
  }
  return 0;
}
```

“data refers to a block of 10 integer variables, data[0], data[1], …, data[9]

The result

Array size should be a constant

```
int main()
{
  int i;
  int data[10];
  for (i=0; i<10; i++) data[i]= i;
  i=0;
  while (i<10)
  {
    printf("Data[%d] = %d\n", i, data[i]);
    i++;
  }
  return 0;
}
```

Output

```
Data[0] = 0
Data[1] = 1
Data[2] = 2
Data[3] = 3
Data[4] = 4
Data[5] = 5
Data[6] = 6
Data[7] = 7
Data[8] = 8
Data[9] = 9
```

Declaring Arrays

- Like variables, the arrays used in a program must be declared before they are used
- General syntax:

  type   array-name [size];

  - type specifies the type of element that will be contained in the array (int, float, char, etc.)
  - size is an integer constant which indicates the maximum number of elements that can be stored inside the array

  int   marks[5];

  - marks is an array that can store a maximum of 5 integers

- Examples:

```
int  x[10];
char  line[80];
float  points[150];
char  name[35];
```

- If we are not sure of the exact size of the array, we can define an array of a large size

```
int   marks[50];
```

though in a particular run we may only be using, say, 10 elements

Accessing Array Elements

- A particular element of the array can be accessed by specifying two things:
  - Name of the array
  - Index (relative position) of the element in the array
- Important to remember: In C, the index of an array **starts from 0**, not 1
- Example:
  - An array is defined as `int x[10];`
  - The first element of the array x can be accessed as x[0], fourth element as x[3], tenth element as x[9], etc.

Contd.

- The array index can be any expression that evaluates to an integer between 0 and n-1 where n is the maximum number of elements possible in the array

```
a[x+2] = 25;
b[3*x-y] = a[10-x] + 5;
```

- Remember that each array element is a variable in itself, and can be used anywhere a variable can be used (in expressions, assignments, conditions,…)

How is an array stored in memory?

- Starting from a given memory location, the successive array elements are allocated space in consecutive memory locations

Array A

  - x: starting address of the array in memory
  - k: number of bytes allocated per array element
- A[i] ➔ is allocated memory location at address x + i*k

A Special Operator: AddressOf (&)

- Remember that each variable is stored at a memory location with an unique address
- Putting & before a variable name gives the starting address of the variable (where it is stored, not the value)
- Can be put before any variable (with no blank in between)

```
        int a =10;
        printf("Value of a is %d, and address of a is
%d\n", a, &a);
```

Example

```
int main()
{
  int i;
  int data[10];
  for(i=0; i<10; i++)
  printf("&Data[%d] = %u\n", i, &data[i]);
  return 0;
}
```

Output

```
&Data[0] = 3221224480
&Data[1] = 3221224484
&Data[2] = 3221224488
&Data[3] = 3221224492
&Data[4] = 3221224496
&Data[5] = 3221224500
&Data[6] = 3221224504
&Data[7] = 3221224508
&Data[8] = 3221224512
&Data[9] = 3221224516
```

Initialization of Arrays

- General form:

  type   array_name[size] = { list of values };

- Examples:

  int marks[5] = {72, 83, 65, 80, 76};

  char name[4] = {‘A’, ‘m’, ‘i’, ‘t’};

- The size may be omitted. In such cases the compiler automatically allocates enough space for all initialized elements

  int   flag[ ] = {1, 1, 1, 0};

  char name[ ] = {‘A’, ‘m’, ‘i’, ‘t’};

How to read the elements of an array?

- By reading them one element at a time

```
for (j=0; j<25; j++)
    scanf ("%f", &a[j]);
```

- The ampersand (&) is necessary
- The elements can be entered all in one line or in different lines

A Warning

- In C, while accessing array elements, array bounds are not checked
- Example:

```
int   marks[5];
:
:
marks[8] = 75;
```

  - The above assignment would not necessarily cause an error
  - Rather, it may result in unpredictable program results, which are very hard to debug

Reading into an array

```
int main()  {
    const int MAX_SIZE = 100;
    int i, size;
    float marks[MAX_SIZE];
    float total;
    scanf("%d",&size);
    for (i=0, total=0; i<size; i++)
    {
        scanf("%f",&marks[i]);
        total = total + marks[i];
     }
   printf("Total = %f \n Avg = %f\n", total,
total/size);
   return 0;
}
```

Output

```
4
2.5
3.5
4.5
5
Total = 15.500000
 Avg = 3.875000
```

How to print the elements of an array?

- By printing them one element at a time

```
for  (j=0; j<25; j++)
    printf  ("\n %f", a[j]);
```

  - The elements are printed one per line

```
printf  ("\n");
for  (j=0; j<25; j++)
    printf (" %f", a[j]);
```

  - The elements are printed all in one line (starting with a new line)

How to copy the elements of one array to another?

- By copying individual elements

```
for  (j=0; j<25; j++)
    a[j] = b[j];
```

- The element assignments will follow the rules of assignment expressions
- Destination array must have sufficient size

Example 1: Find the minimum of a set of 10 numbers

```
int main()
{
    int  a[10], i, min;

    for  (i=0; i<10; i++)
        scanf (“%d”, &a[i]);

    min = a[0];
    for  (i=1; i<10; i++)
    {
        if  (a[i] < min)
            min = a[i];
    }
    printf (“\n Minimum is %d”, min);
    return 0;
}
```

Alternate Version 1

Change only one line to change the problem size

```
const int size = 10;

int main()
{
    int  a[size], i, min;

    for  (i=0; i<size; i++)
        scanf (“%d”, &a[i]);

    min = a[0];
    for  (i=1; i<size; i++)
    {
        if  (a[i] < min)
            min = a[i];
    }
    printf (“\n Minimum is %d”, min);
    return 0;
}
```

Alternate Version 2

```
#define size 10

int main()
{
    int  a[size], i, min;

    for  (i=0; i<size; i++)
        scanf ("%d", &a[i]);

    min = a[0];
    for  (i=1; i<size; i++)
    {
        if  (a[i] < min)
            min = a[i];
    }
    printf ("\n Minimum is %d", min);
    return 0;
}
```

Change only one line to change the problem size

Used #define macro

#define macro

- #define X Y
- Preprocessor directive
  - The #include you have been using is also a preprocessor directive
- Compiler will first replace all occurrences of string X with string Y in the program, then compile the program
- Similar effect as read-only variables (const), but no storage allocated

Alternate Version 3

Define an array of large size and use only the required number of elements

```
int main()
{
   int  a[100], i, min, n;

   scanf ("%d", &n);  /* Number of elements */
   for  (i=0; i<n; i++)
      scanf ("%d", &a[i]);

   min = a[0];
   for  (i=1; i<n; i++)
   {
      if  (a[i] < min)
         min = a[i];
   }
   printf ("\n Minimum is %d", min);
   return 0;
}
```

Example 2: Computing cgpa

Handling two arrays at the same time

```
const int  nsub = 6;

int main()
{
   int  grade_pt[nsub], cred[nsub], i, gp_sum=0,
cred_sum=0;
   double gpa;

   for  (i=0; i<nsub; i++)
      scanf (“%d %d”, &grade_pt[i], &cred[i]);

   for  (i=0; i<nsub; i++)
   {
      gp_sum += grade_pt[i] * cred[i];
      cred_sum += cred[i];
   }
   gpa = ((float) gp_sum) / cred_sum;
   printf (“\n Grade point average:  is %.2lf”, gpa);
   return 0;
}
```

Things you cannot do

- You cannot
  - use = to assign one array variable to another

    a = b;   /* a and b are arrays */
  - use == to directly compare array variables

    if (a = = b) ………..
  - directly scanf or printf arrays

    printf (“……”, a);